An exploration on differentiability and continuity in relation to difference quotient

What is differentiability?

Differentiability is a fundamental concept in calculus that describes whether a function has a well-defined derivative at a given point. Although analytical methods can establish differentiability through limits and derivative rules, numerical methods provide an alternative approach for investigating the behavior of functions. However, the main question is whether these numerical methods can accurately estimate differentiability at a given point while considering the continuity of a function. To investigate this question, I used AI assistance and my understanding of the numerical method known as the difference quotient to develop an interactive application. This application allows users to estimate and investigate differentiability while considering the mathematical conditions required for a function to be differentiable, including continuity that requires the absence of sharp corners, cusps, jumps, vertical tangents. Through this investigation, I aim to demonstrate how numerical approximations can be used to connect theories from calculus concepts with computational methods. Within the app there will be functions with different differentiability characteristics which will allow for the observation by the user to see how step sizes for the estimations of derivatives affect the accuracy of the estimation. The app will also have the rationality to indicate if the point is differentiable or not using the prementioned restrictions of differentiable points.

Mathematical theory

To understand our reasoning behind the code, let us explore some of the mathematical theory behind derivatives and how it relates to the difference quotient. The formal definition of a derivative is:

$$f'(x) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

Here, a represents the point of evaluation, while h represents the step size, or horizontal distance between the two points. As h approaches zero from both the left and right, the difference quotient approaches the exact derivative, provided that the limit exists and is finite. In numerical applications, we instead use a small, nonzero value of h to estimate this derivative. Additionally, considering that the derivative represents the slope of a tangent line, we can see that as h approaches zero, the secant slope approaches the tangent line's slope.

To determine whether a function is differentiable at a selected point, we must compare the left- and right-hand difference quotients. This follows from the properties of limits: if the left- and right-hand limits approach different values, then the two-sided limit does not exist. Therefore, the function is not differentiable at that point.

With this in mind, we must use the following formulas of the left and right difference quotients:

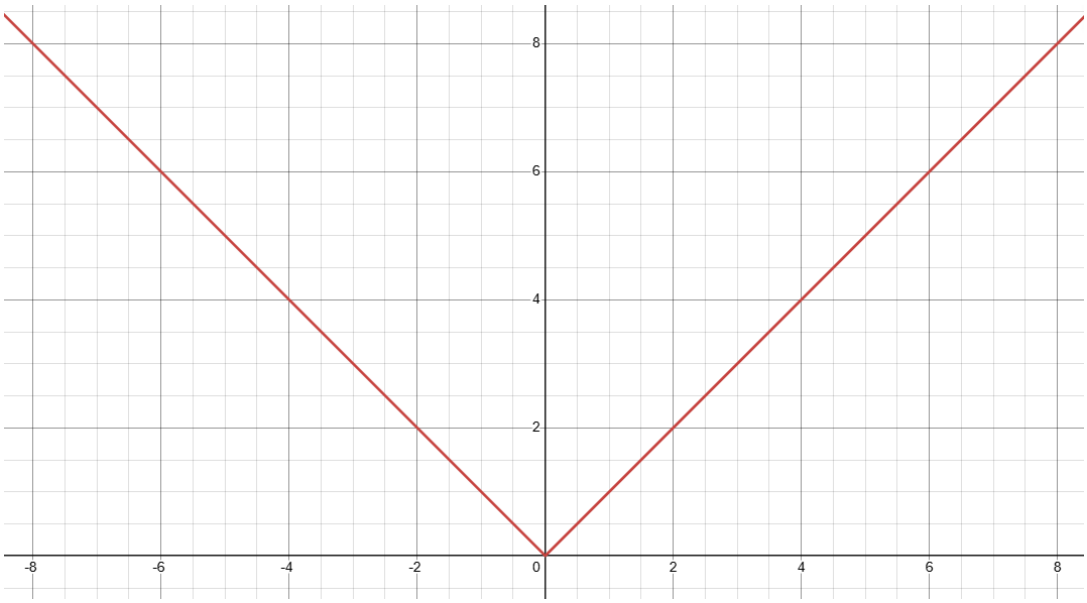
$$D_R(a,h) = \frac{f(a+h) - f(a)}{h} \;\; \& \;\; D_L(a,h) = \frac{f(a) - f(a-h)}{h}$$

Using these equations and the theory of limits established in the previous paragraph, we can conclude that a function is differentiable at a point if both one-sided difference quotients approach the same finite value as the step size approaches zero:

$$D_R(a,h) \;=\; D_L(a,h)$$

This is a fundamental mathematical condition used within the application's code to assess whether differentiability is likely at any selected point “a”.

One final mathematical concept to consider when using numerical approximations of derivatives is continuity and how differentiability depends upon it. Continuity at a given point is required for that point to be differentiable. This can be related to our previous discussion of limits, since continuity requires the left and right hand limits of the function to approach the same value. However, being continuous does not necessarily guarantee that a function is differentiable at that point. An example of a continuous function with a non-differentiable point would be f(x) = |x| (figure a):



(Figure a)

Letting our point of evaluation be a = 0, we observe a sharp corner where there is no single, well-defined tangent slope. Using numerical approximations, we can investigate why this point is not differentiable. Applying our previous discussion of limits to the difference quotients, we find that the left-hand derivative approaches -1, while the right-hand derivative approaches +1. Since these values do not agree, the two-sided derivative limit does not exist, indicating that the function is nondifferentiable at a = 0.

Numerical Method used in app

Using the theory that was presented in the previous section, let us now apply it to formulate our logical reasoning within the main application of the differentiability app. To start, let us use the fact that our left and right numerical estimates of difference quotients should agree numerically for a point to be differentiable. Within the app to achieve a reasonable evaluation of a differentiable point, we set the tolerance of stating likely differentiable to 0.01:

$$|D_R(a,h) - D_L(a,h)| \leq 0.01$$

This means that if there is an absolute difference greater than 0.01 of the two numerical estimates then the point is likely not differentiable. To allow for better accuracy of the secant line as h approaches 0, we used 0.001 as our minimum value of h with a step size of 0.01 to allow for greater restrictions on the statement of a “likely differentiable point.”

We must also consider the possible chance of an undefined, vertical slope. To account for this, we used:

$$|D_L| > 50, |D_R| > 50, and\ sign(D_L) = sign(D_R)$$

This condition checks whether the absolute values of both derivative estimates exceed a specified threshold and whether both estimates have the same sign. If these conditions are satisfied, the application identifies a possible vertical tangent. However, this classification is only an approximation because a large numerical slope does not necessarily indicate an undefined derivative.

In addition to the mathematical theory and interactive sliders, the application contains a graphing window with four predefined functions: x^2, $|x|$, $\sqrt[3]{x}$, and a piecewise function with a jump discontinuity. These functions allow users to investigate different examples of differentiability, non-differentiability, and continuity through graphical and numerical observations. Our application also contains sliders for x axis domain, h, and a that indicate the slopes of the secant and tangent lines instantly as they are moved along the selected graph.

Results and Interpretation

For this section we will conduct an example experiment using the app. This will be exploring a differentiable everywhere graph, x^2, and analyzing our estimates of left and right to see how close our estimate gets. We will use the following settings:

F(x)= x^2

h = 0.001

a = 2

along with the know value of F'(x) at 2 (which is 4) to compare the 2 estimates which are

$$D_R(2,0.001) = 4.001\ \ \&\ \ D_L(2,0.001) = 3.999$$

Using our tolerance inequality we see that:

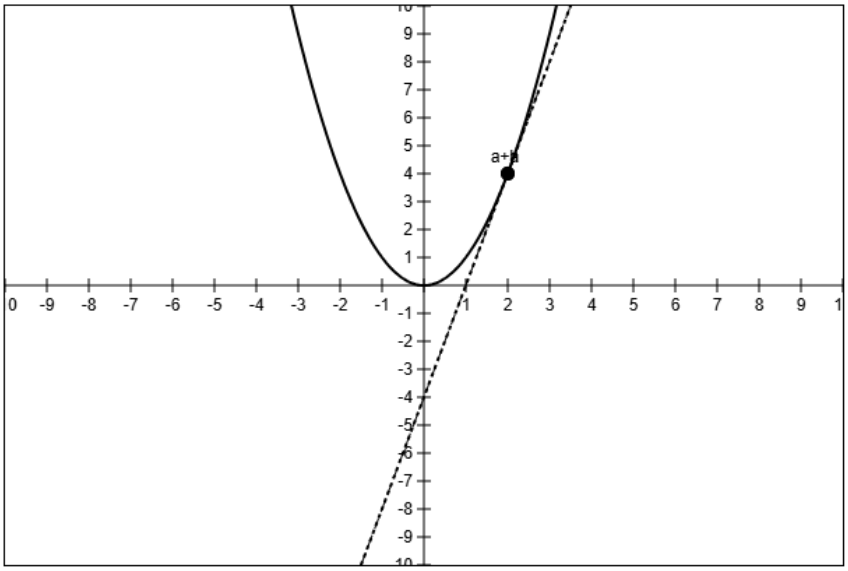
$$|4.001 - 3.999| = 0.002$$

Which is less than our tolerance value of 0.01 indicating that this point is potentially differentiable

Smooth Function x² Submit

Your selection

f(x) = x²



Graph Range

Minimum x: -10

Maximum x: 10

Point a: 2

Increment h: 0.001

Show Secant Line
Show Tangent Line

Secant slope [f(a+h) - f(a)] / h: 4.0010

Numerical Diagnostics

f(a): 4.0000

f(a+h): 4.0040

Left derivative estimate: 3.9990

Right derivative estimate: 4.0010

Tangent slope estimate: 4.0000

Conclusion: Likely differentiable at a